

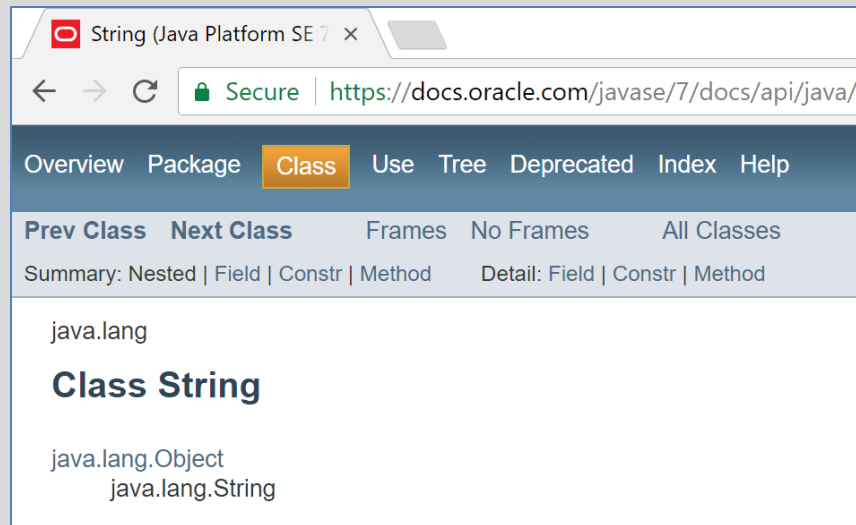
Recap of OO concepts

Objects, classes, methods and more.

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Classes and Objects

- A **class**
 - defines a group of related **methods** (functions) and **fields** (variables / properties).

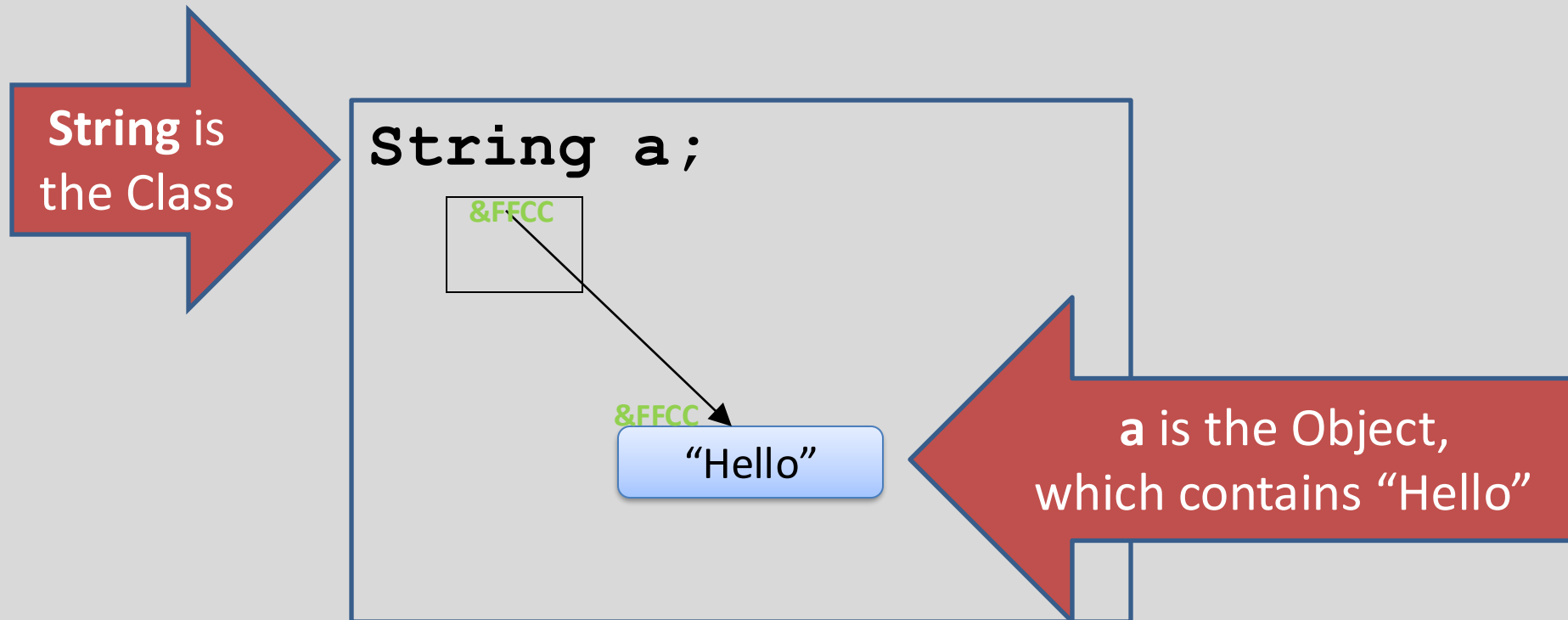


The screenshot shows the Java API documentation for the `String` class in the `java.lang` package. The browser address bar shows the URL `https://docs.oracle.com/javase/7/docs/api/java/`. The navigation bar includes links for Overview, Package, Class (highlighted), Use, Tree, Deprecated, Index, and Help. Below the navigation bar, there are links for Prev Class, Next Class, Frames, No Frames, and All Classes. The main content area shows the package `java.lang` and the class `String`, with a hierarchy showing `java.lang.Object` and `java.lang.String`.

Method Summary	
Methods	
Modifier and Type	Method and Description
char	<code>charAt(int index)</code> Returns the char value at the specified index.
int	<code>codePointAt(int index)</code> Returns the character (Unicode code point) at the specified index.
int	<code>codePointBefore(int index)</code> Returns the character (Unicode code point) before the specified index.
int	<code>codePointCount(int beginIndex, int endIndex)</code> Returns the number of Unicode code points in the specified text range of this <code>String</code> .
int	<code>compareTo(String anotherString)</code> Compares two strings lexicographically.
int	<code>compareToIgnoreCase(String str)</code> Compares two strings lexicographically, ignoring case differences.
<code>String</code>	<code>concat(String str)</code> Concatenates the specified string to the end of this string.
boolean	<code>contains(CharSequence s)</code> Returns true if and only if this string contains the specified sequence of char values.
boolean	<code>contentEquals(CharSequence cs)</code> Compares this string to the specified <code>CharSequence</code> .
boolean	<code>contentEquals(StringBuffer sb)</code> Compares this string to the specified <code>StringBuffer</code> .
static <code>String</code>	<code>copyValueOf(char[] data)</code> Returns a <code>String</code> that represents the character sequence in the array specified.
static <code>String</code>	<code>copyValueOf(char[] data, int offset, int count)</code> Returns a <code>String</code> that represents the character sequence in the array specified.
boolean	<code>endsWith(String suffix)</code> Tests if this string ends with the specified suffix.
boolean	<code>equals(Object anObject)</code> Compares this string to the specified object.

Classes and Objects

- An **object**
 - is a single instance of a class
 - i.e. an object is created (instantiated) from a class.



Classes and Objects – Many Objects

- Many **objects** can be constructed from a single **class** definition.
- Each **object** must have a unique name within the program.

Ver 1.0

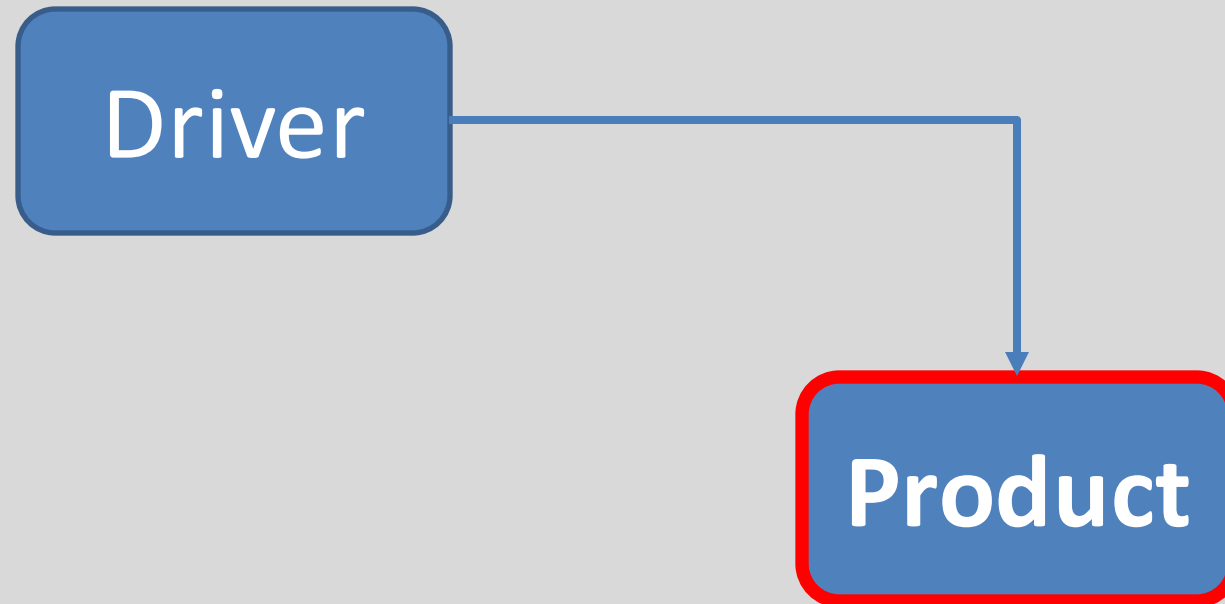
SHOP



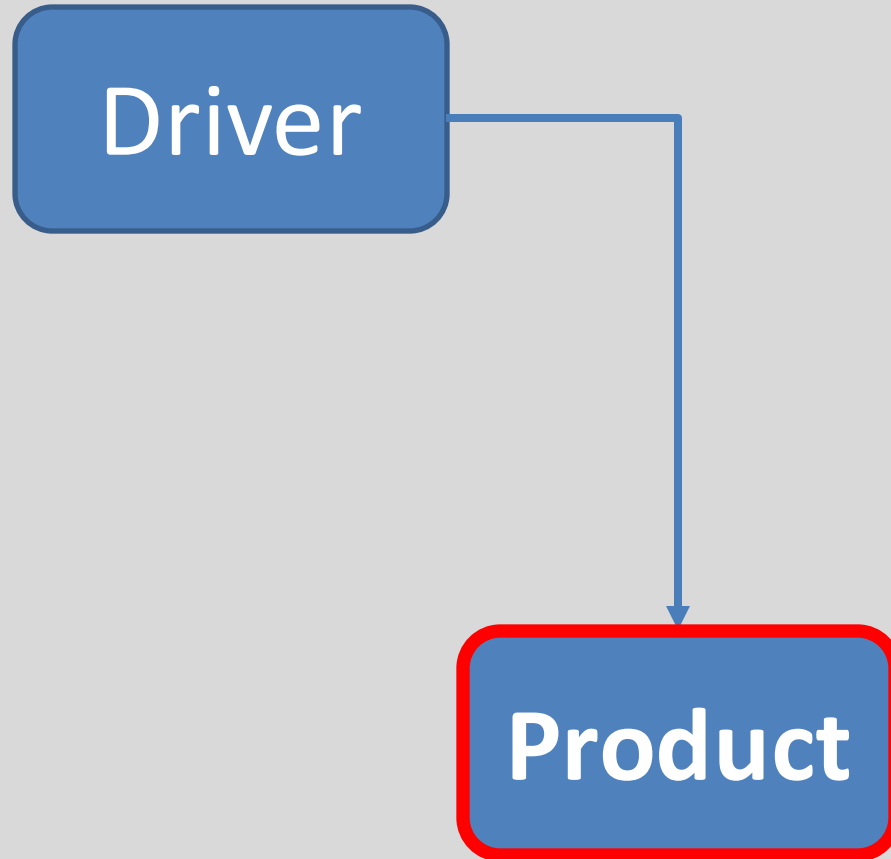
Shop V1.0 - Product



- We will recap object oriented concepts through the study of a new class called **Product**.



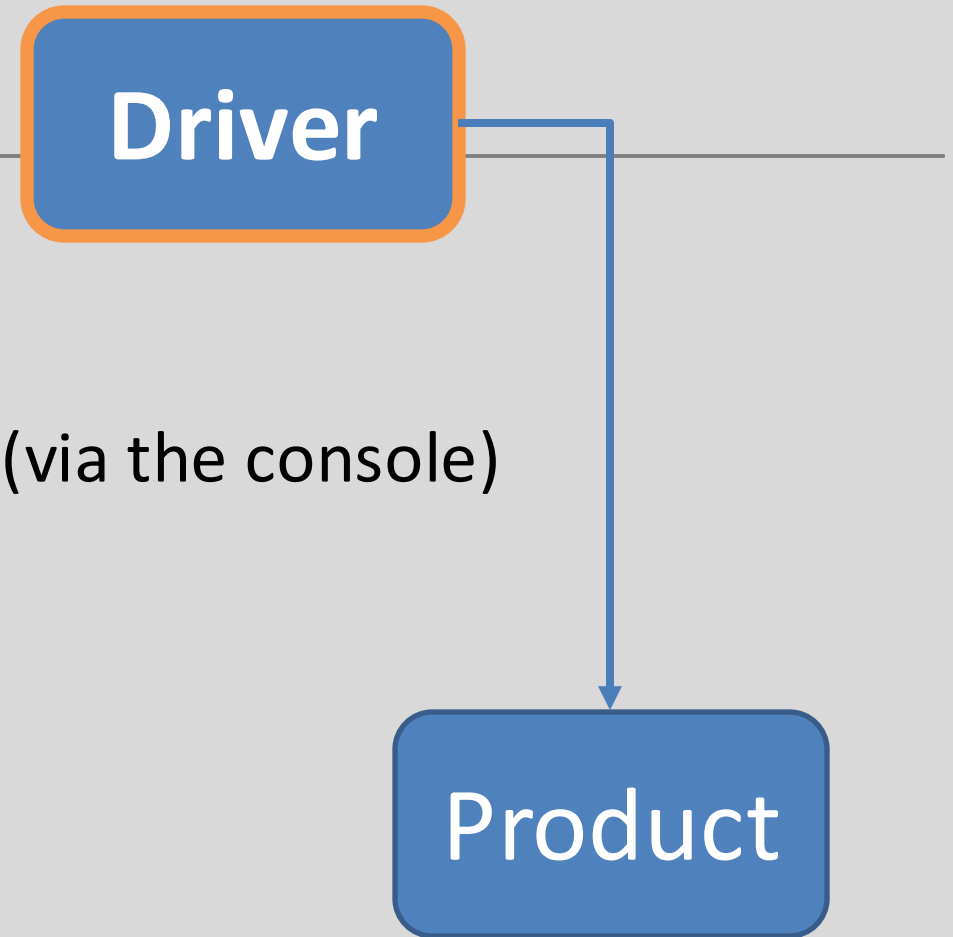
Shop V1.0 - Product



- The **Product** class stores **details** about a product
 - name
 - code
 - unit cost
 - in the current product line or not?

Shop V1.0 - Driver

- The **Driver** class
 - has the **main()** method.
 - **reads** the product details from the user (via the console)
 - **creates** a new Product object.
 - **prints** the product object (to the console)
- **Driver** is covered in the next lecture.



A Product Class...

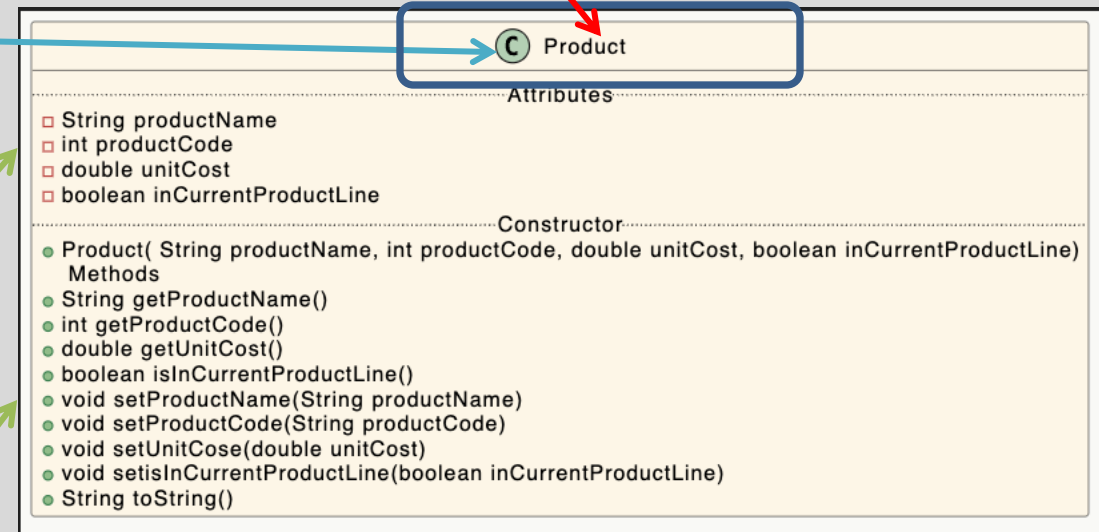
Object Type/ **Class** Name
i.e. Product



The **C** icon means it is a **Class**.

The red box means it is **private**.

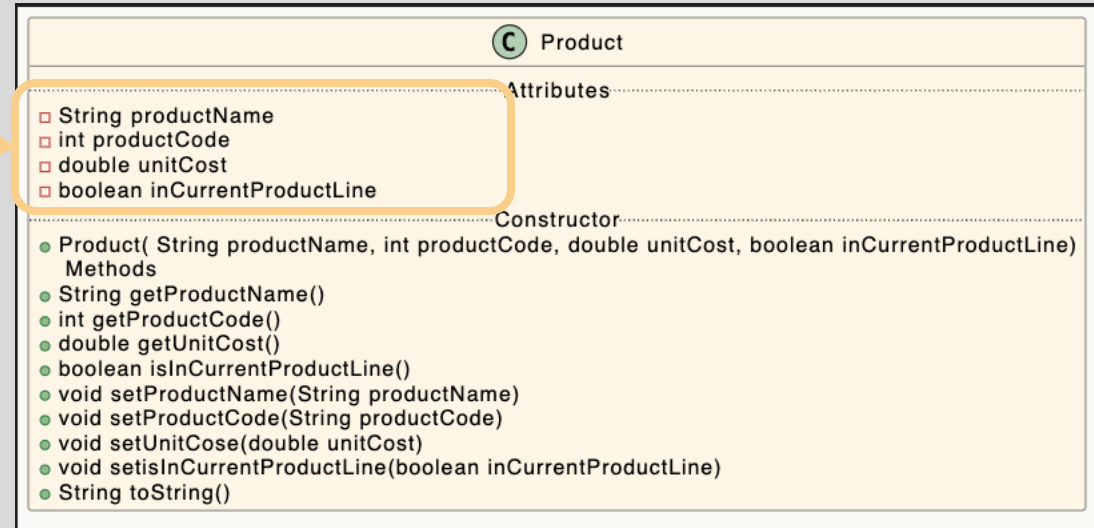
The green circle means it is **public**.



A Product Class...fields

Fields

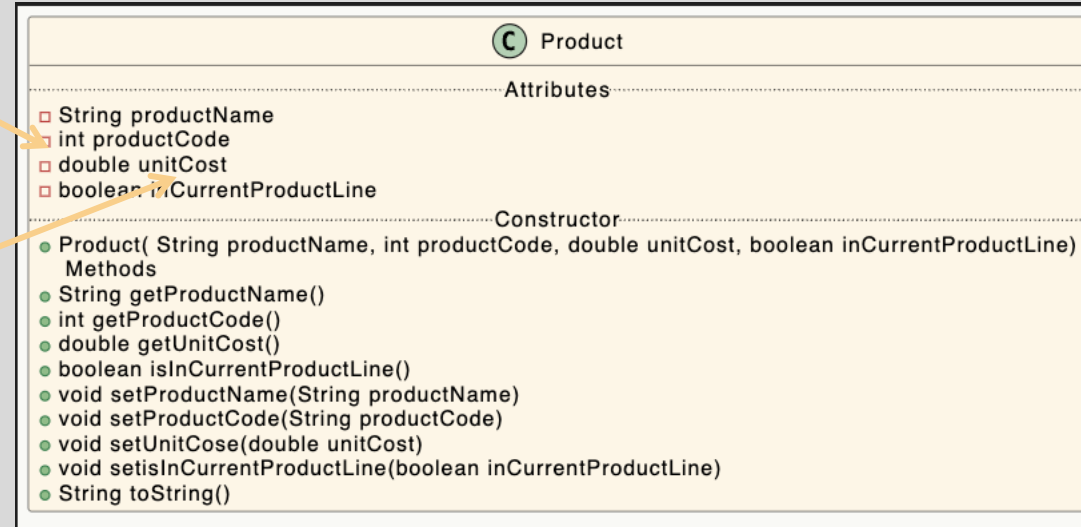
i.e. the **attributes / properties** of the class



A Product Class...fields

field **type**

field **name**

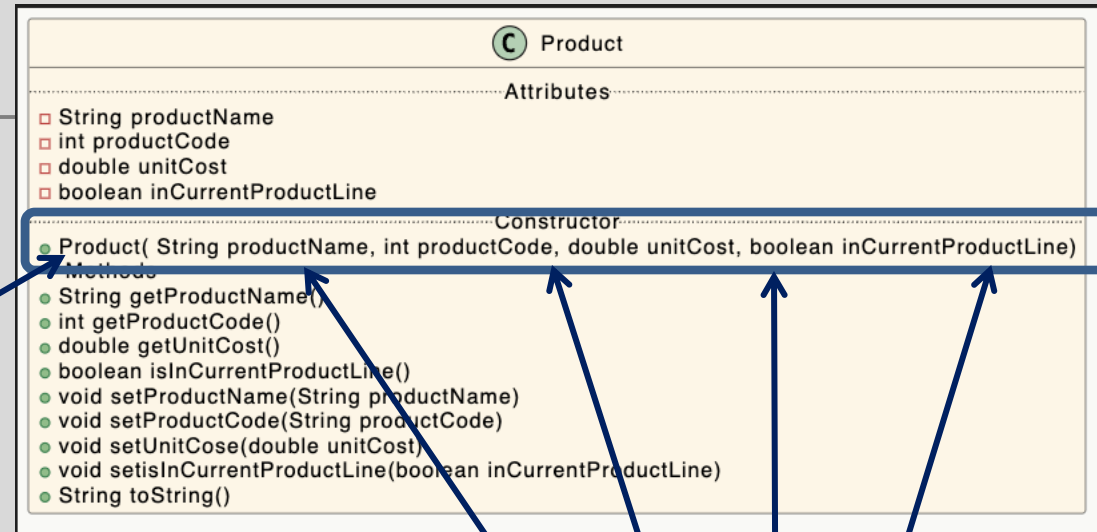


A Product Class...

Constructor

Constructor
i.e. for building objects.

Constructors have same name as the class



Four **parameters**;
one for each field.

A Product Class... fields and constructor

```
public class Product {
```

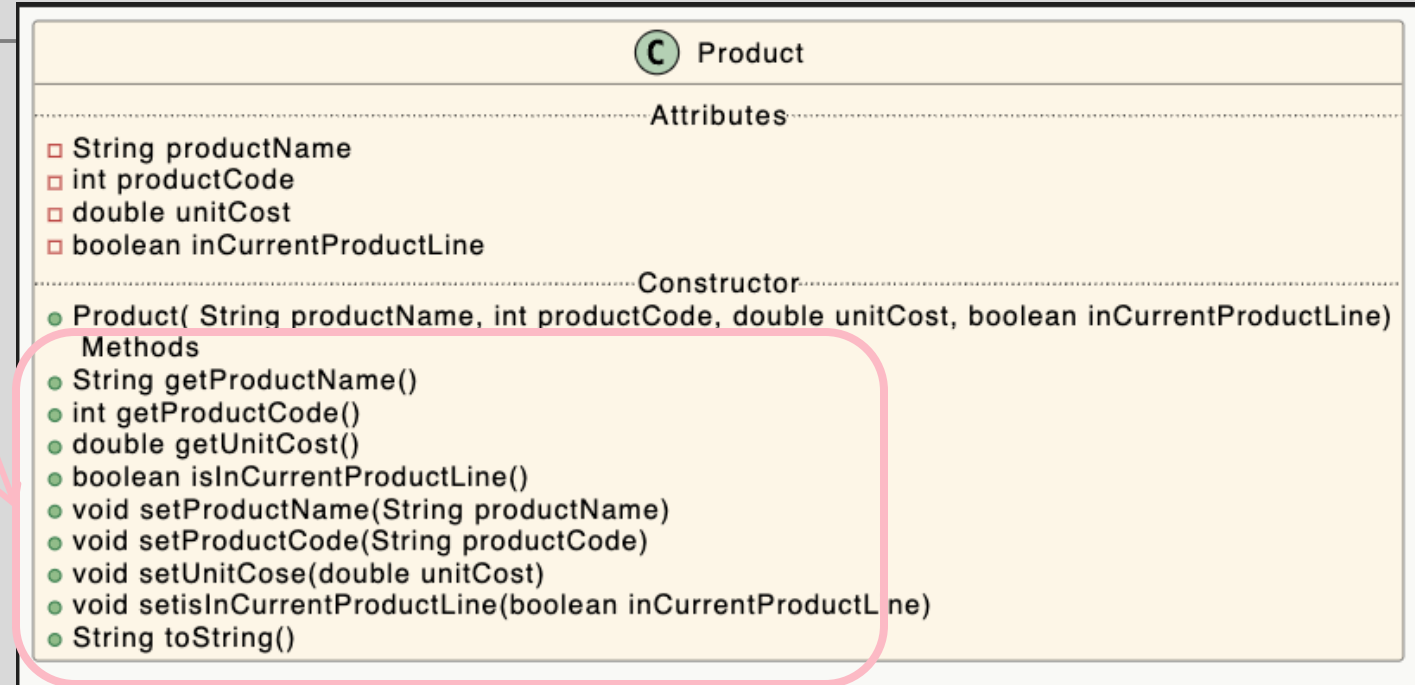
```
    private String productName;  
    private int productCode;  
    private double unitCost;  
    private boolean inCurrentProductLine;
```

```
    public Product (String productName, int productCode,  
                    double unitCost, boolean inCurrentProductLine) {  
  
        this.productName = productName;  
        this.productCode = productCode;  
        this.unitCost = unitCost;  
        this.inCurrentProductLine = inCurrentProductLine;  
    }
```

A Product Class... methods

Methods

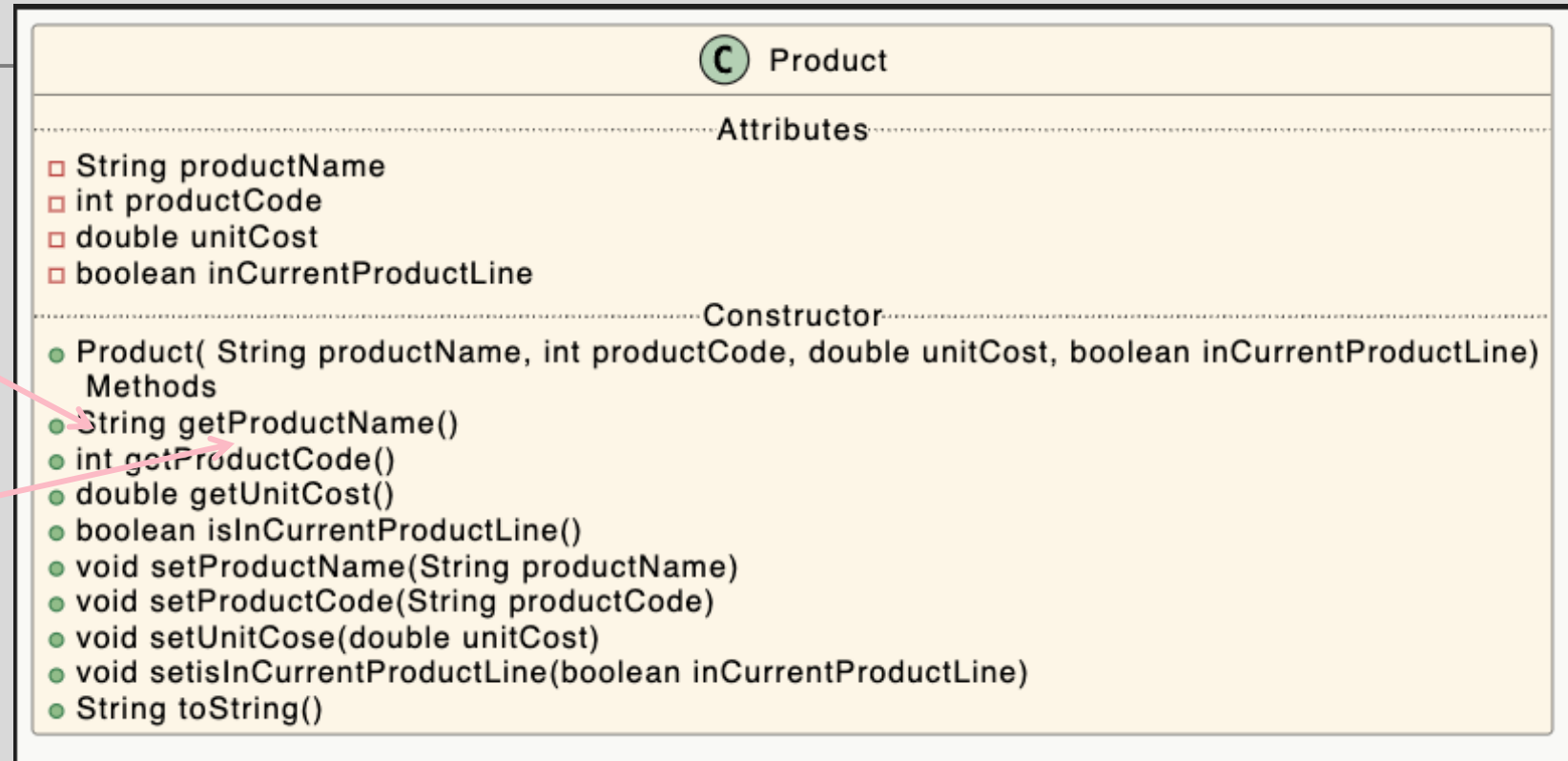
i.e. the **behaviours** of the class



A Product Class... methods

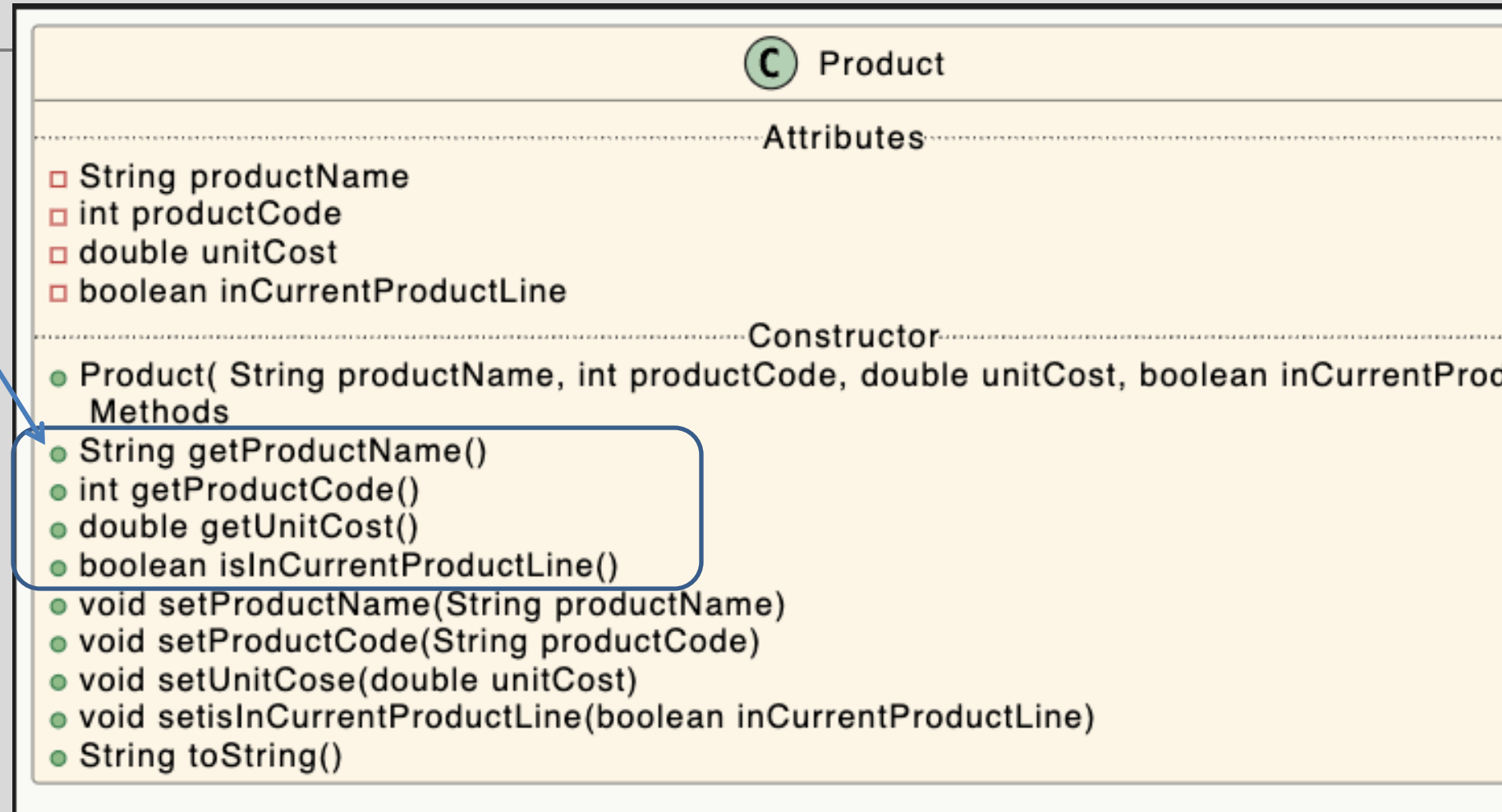
Return type

Method name



A Product Class... **getters**

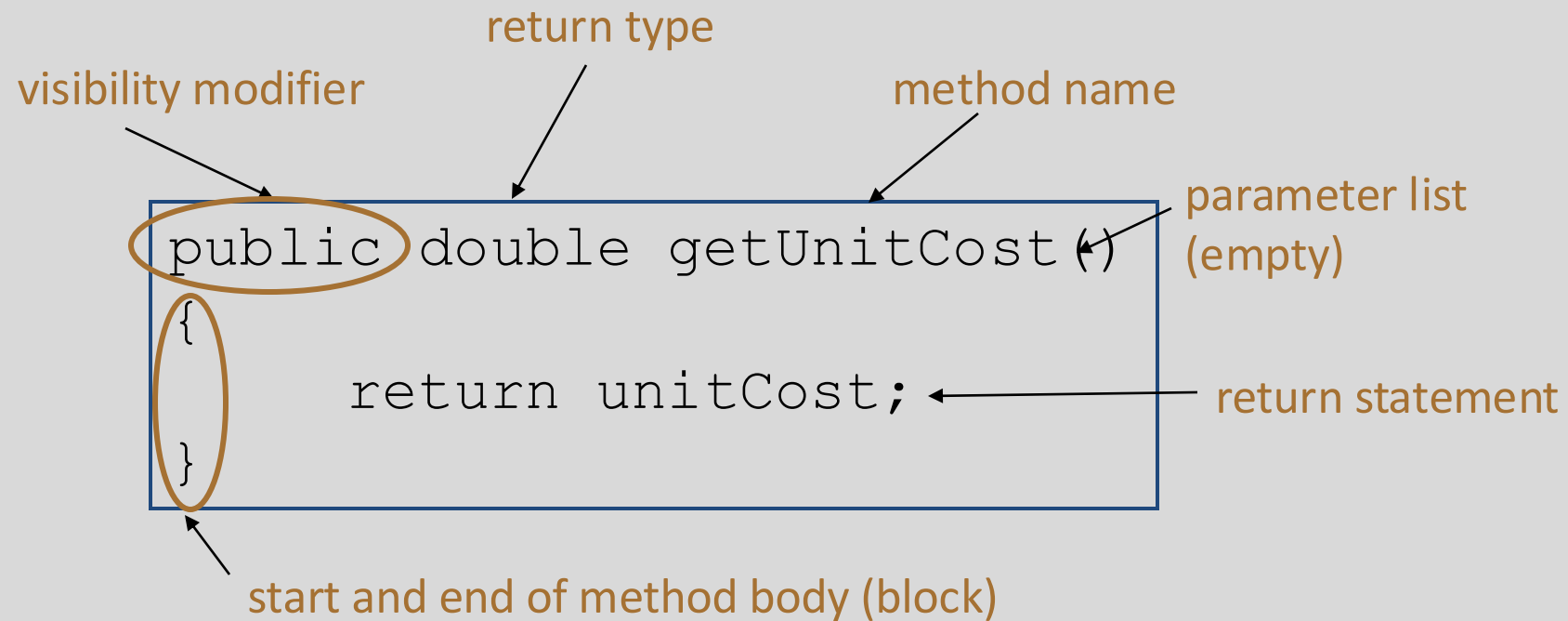
getters



Getters (Accessor Methods)

- **Accessor** methods
 - return information about the **state** of an object
 - i.e. **the values stored in the fields.**
- A **'getter'** method
 - is a specific type of **accessor** method and typically:
 - **contains a return statement**
(as the last executable statement in the method).
 - defines a **return type.**
 - **does NOT change the object state.**

Getters

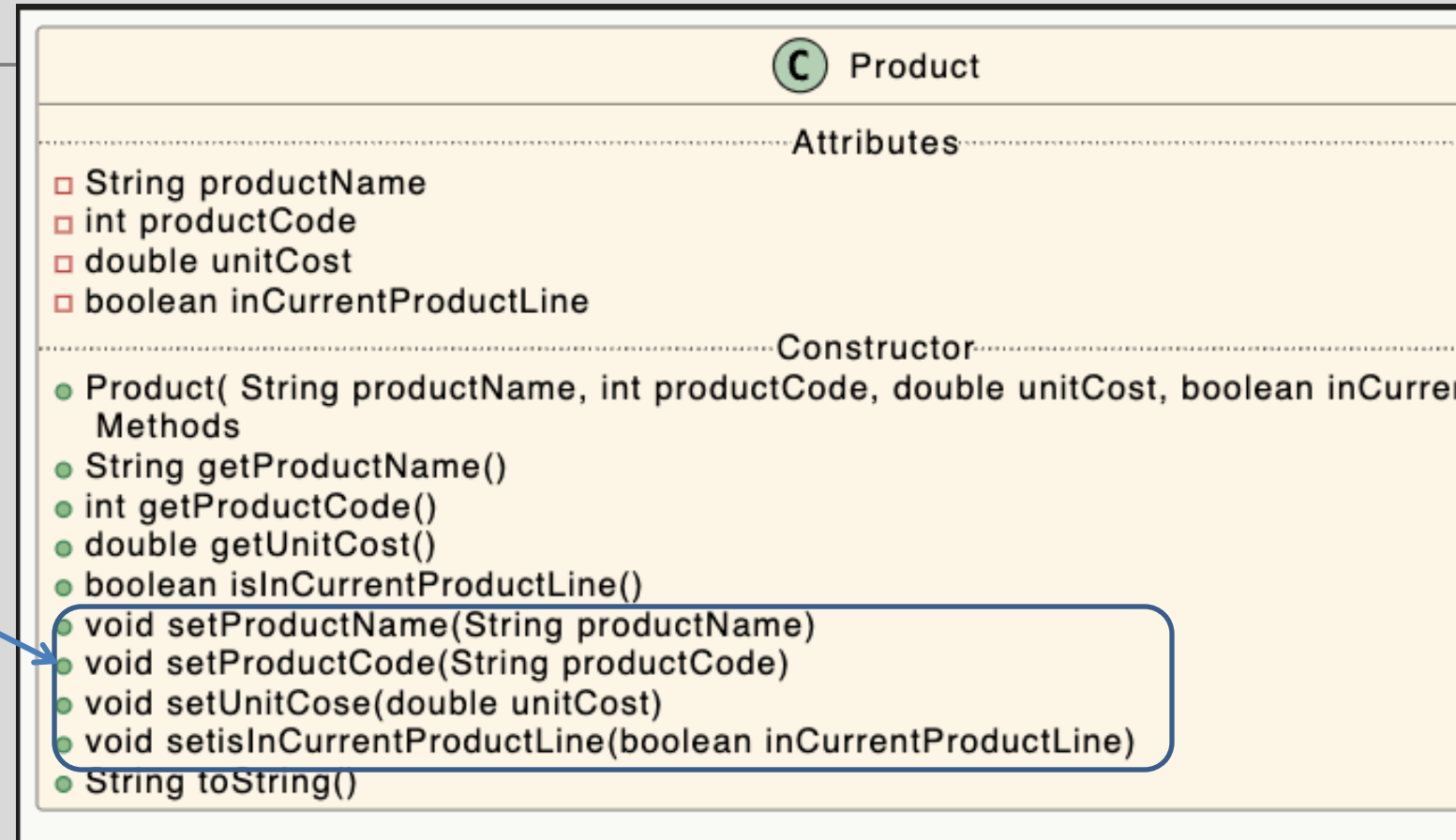


A Product Class...getters

```
public String getProductName() {  
    return productName;  
}  
  
public double getUnitCost() {  
    return unitCost;  
}  
  
public int getProductCode() {  
    return productCode;  
}  
  
public boolean isInCurrentProductLine() {  
    return inCurrentProductLine;  
}
```

A Product Class...setters

setters



Setters (Mutator methods)

- **Mutator** methods
 - change (i.e. mutate!) an object's state.
- A **'setter'** method
 - is a specific type of **mutator** method and typically:
 - contains an **assignment statement**
 - takes in a **parameter**
 - **changes the object state.**

Setters

The diagram illustrates the components of a Java setter method. The code is enclosed in a blue rectangular box. Annotations with arrows point to specific parts of the code:

- visibility modifier**: Points to the word `public`.
- return type**: Points to the word `void`.
- method name**: Points to the text `setUnitCost`.
- parameter**: Points to the text `double unitCost` inside the parentheses.
- field being mutated**: Points to the text `this.unitCost` on the right-hand side of the assignment.
- assignment statement**: Points to the equals sign `=`.
- Value passed as a parameter**: Points to the text `unitCost` on the left-hand side of the assignment.

```
public void setUnitCost(double unitCost)
{
    this.unitCost = unitCost;
}
```

A Product Class...setters

```
public void setProductCode(int productCode) {  
    this.productCode = productCode;  
}  
  
public void setProductName(String productName) {  
    this.productName = productName;  
}  
  
public void setUnitCost(double unitCost) {  
    this.unitCost = unitCost;  
}  
  
public void setCurrentProductLine(boolean inCurrentProductLine) {  
    this.inCurrentProductLine = inCurrentProductLine;  
}
```

Getters/Setters

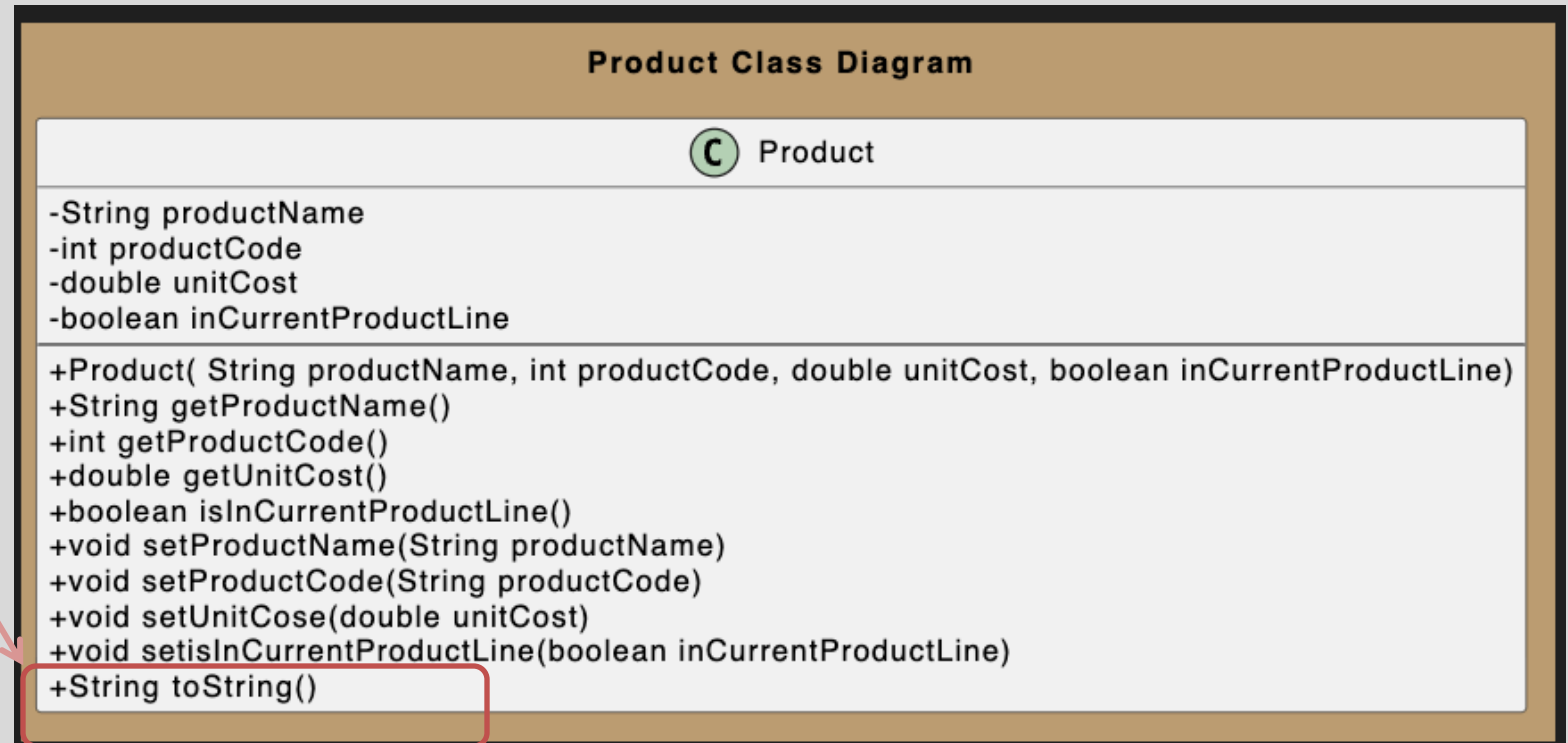
For **each instance field** in a class, you are normally asked to write:

- A **getter**
 - Return statement
- A **setter**
 - Assignment statement

A Product Class...toString

toString():

Builds and returns a String containing a user-friendly representation of the object state.



A Product Class...

```
public String toString()
{
    return "Product description: " + productName
        + ", product code: " + productCode
        + ", unit cost: " + unitCost
        + ", currently in product line: " + inCurrentProductLine;
}
```

Sample Console Output if we printed a Product Object:

Product description: 55 Inch TV, product code: 23432, unit cost: 1399.99, currently in product line: true

toString()

- This is a useful method and you will write a **toString()** method for most of your classes.
 - **When you print an object, Java automatically calls the toString() method** 
- e.g.

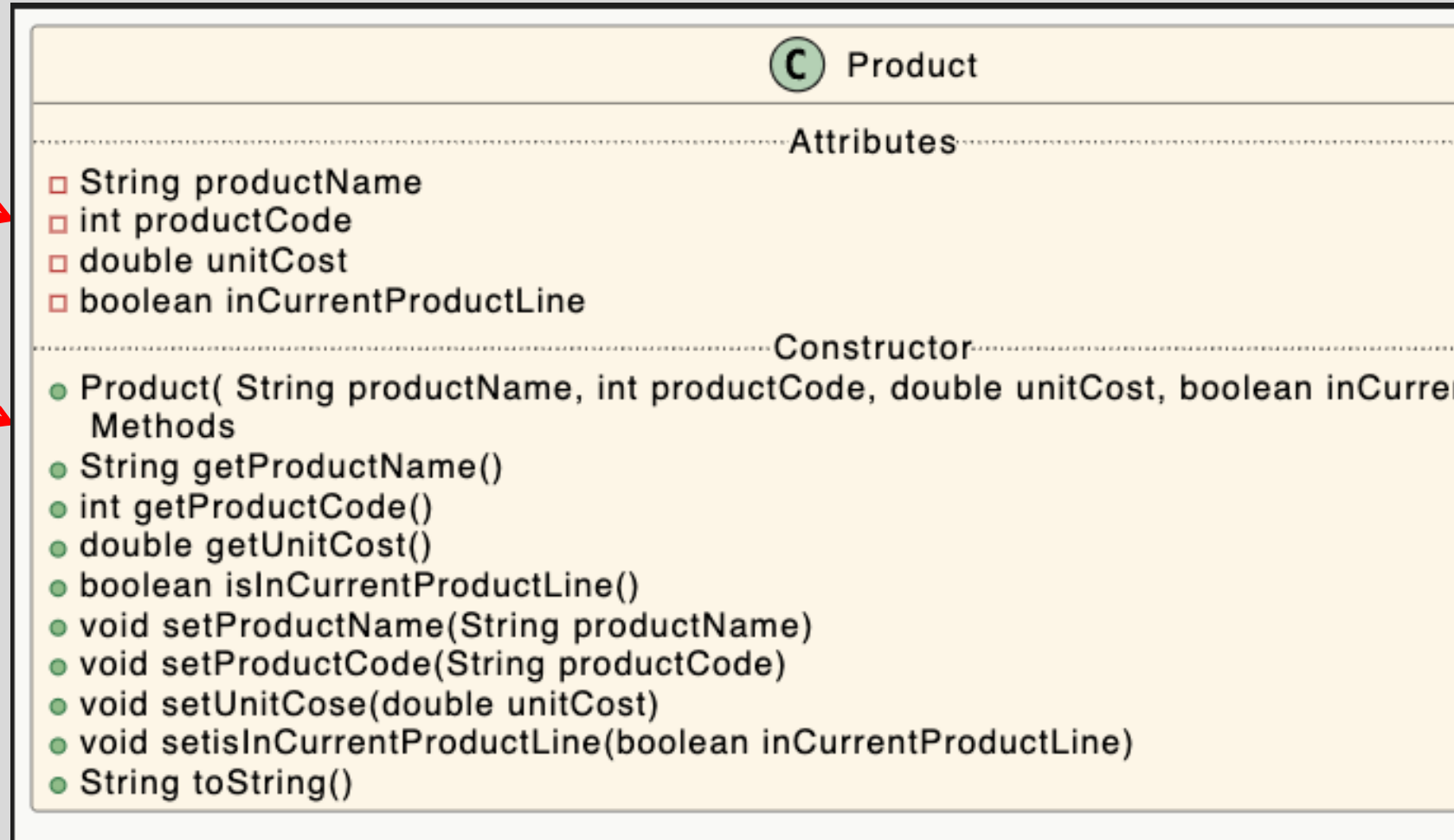
```
Product product = new Product();  
  
//both of these lines of code do the same thing  
System.out.println(product);  
System.out.println(product.toString());
```

Encapsulation in Java – steps 1-3

Encapsulation Step	Approach in Java
1. Wrap the data (fields) and code acting on the data (methods) together as single unit.	<pre>public class <i>ClassName</i> { <i>Fields</i> <i>Constructors</i> <i>Methods</i> }</pre>
2. Hide the fields from other classes.	Declare the fields of a class as <u>private</u>.
3. Access the fields only through the methods of their current class.	Provide <u>public</u> setter and getter methods to modify and view the fields values.

A Product Class... An Encapsulated Class

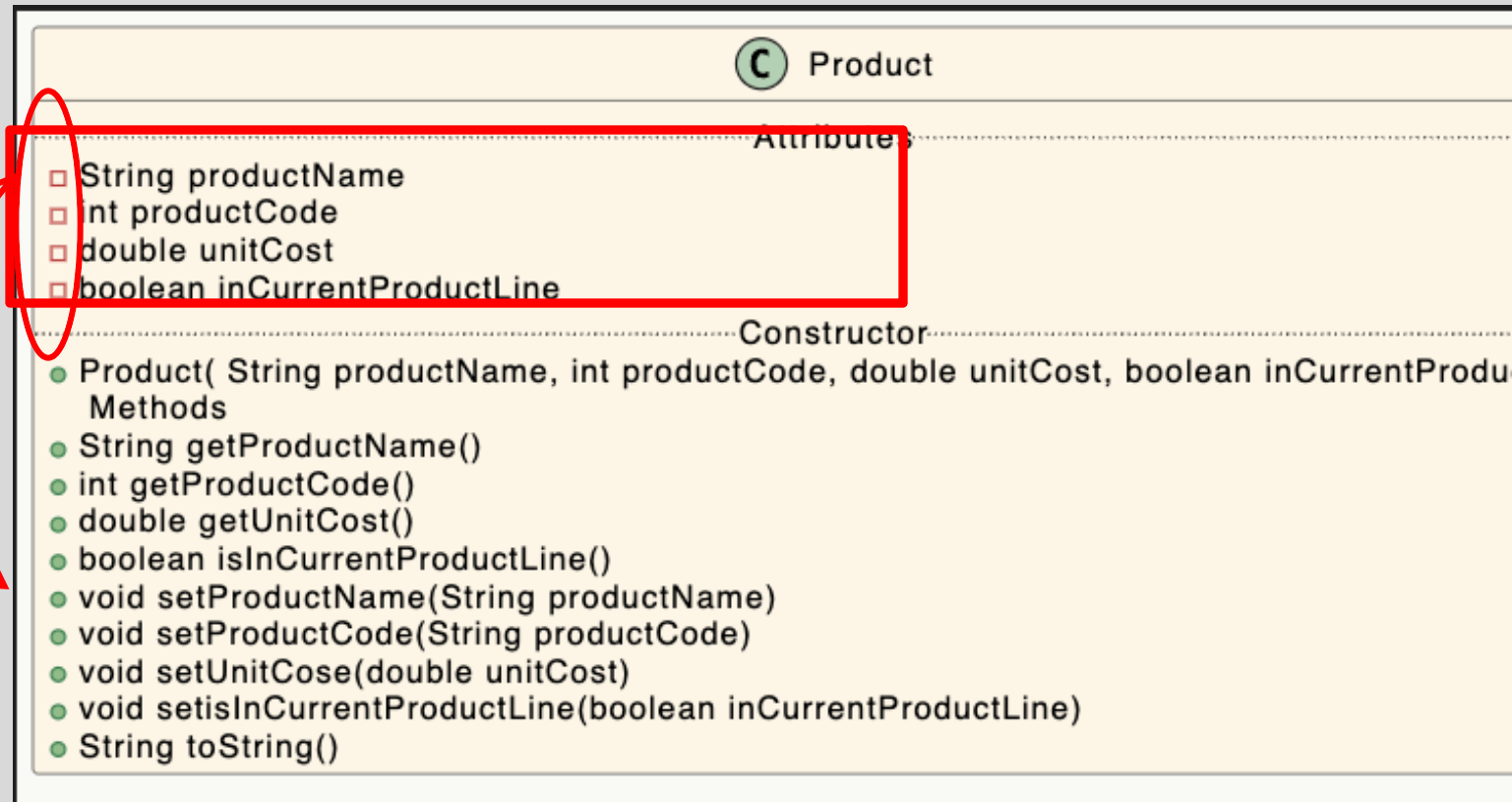
1. Product class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.



A Product Class... An Encapsulated Class

1. Product class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.

2. Fields are **hidden** from other classes.

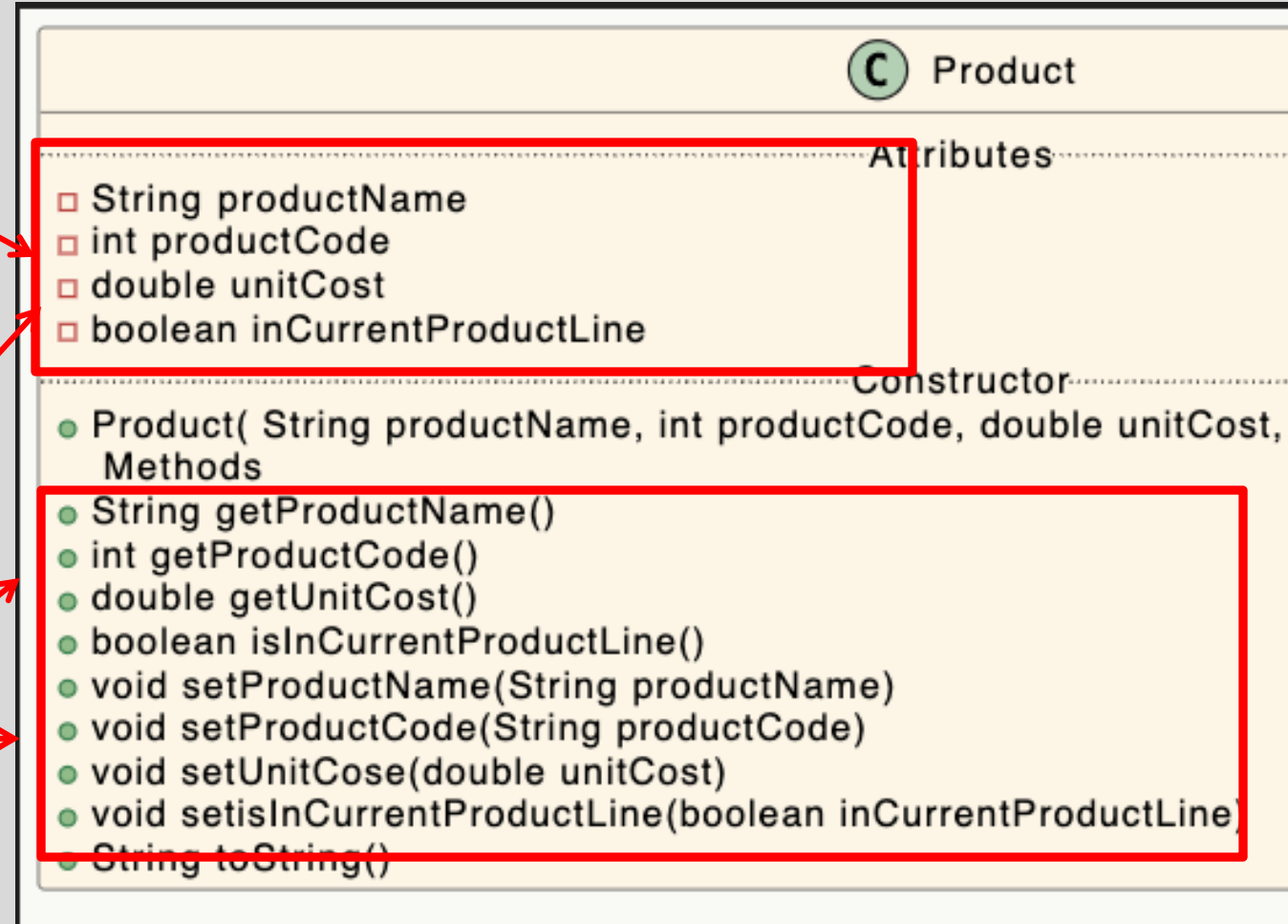


A Product Class... An Encapsulated Class

1. Product class **wraps** the data (fields) and code acting on the data (methods) together as **single unit**.

2. Fields are **hidden** from other classes.

3. **Access** the fields only through the methods of Product (e.g. **getter** and **setter** methods).



Using the Product Class

1

```
private Product product;
```

Declaring an object
product, of type
Product.

product

null

Using the Product Class

1

```
private Product product;
```

Declaring an object **product**, of type **Product**.

product



2

```
product = new Product("TV", 1234, 1149.99, true);
```

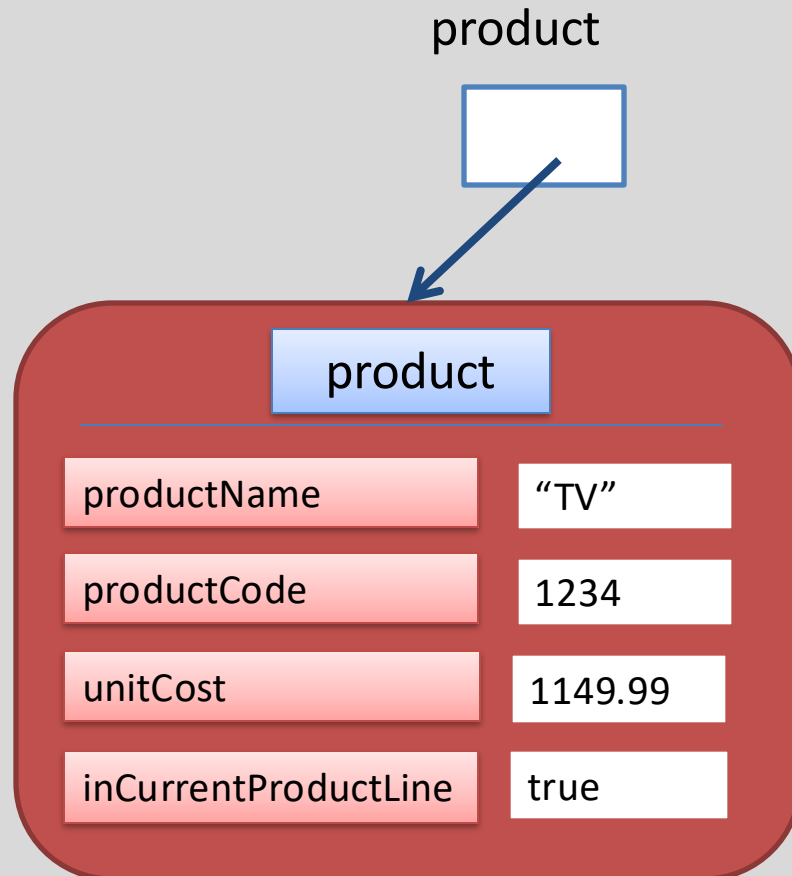
Calls the **Product** *constructor* to build the **product** object in memory.

product

product	
productName	"TV"
productCode	1234
unitCost	1149.99
inCurrentProductLine	true

Multiple Product Objects

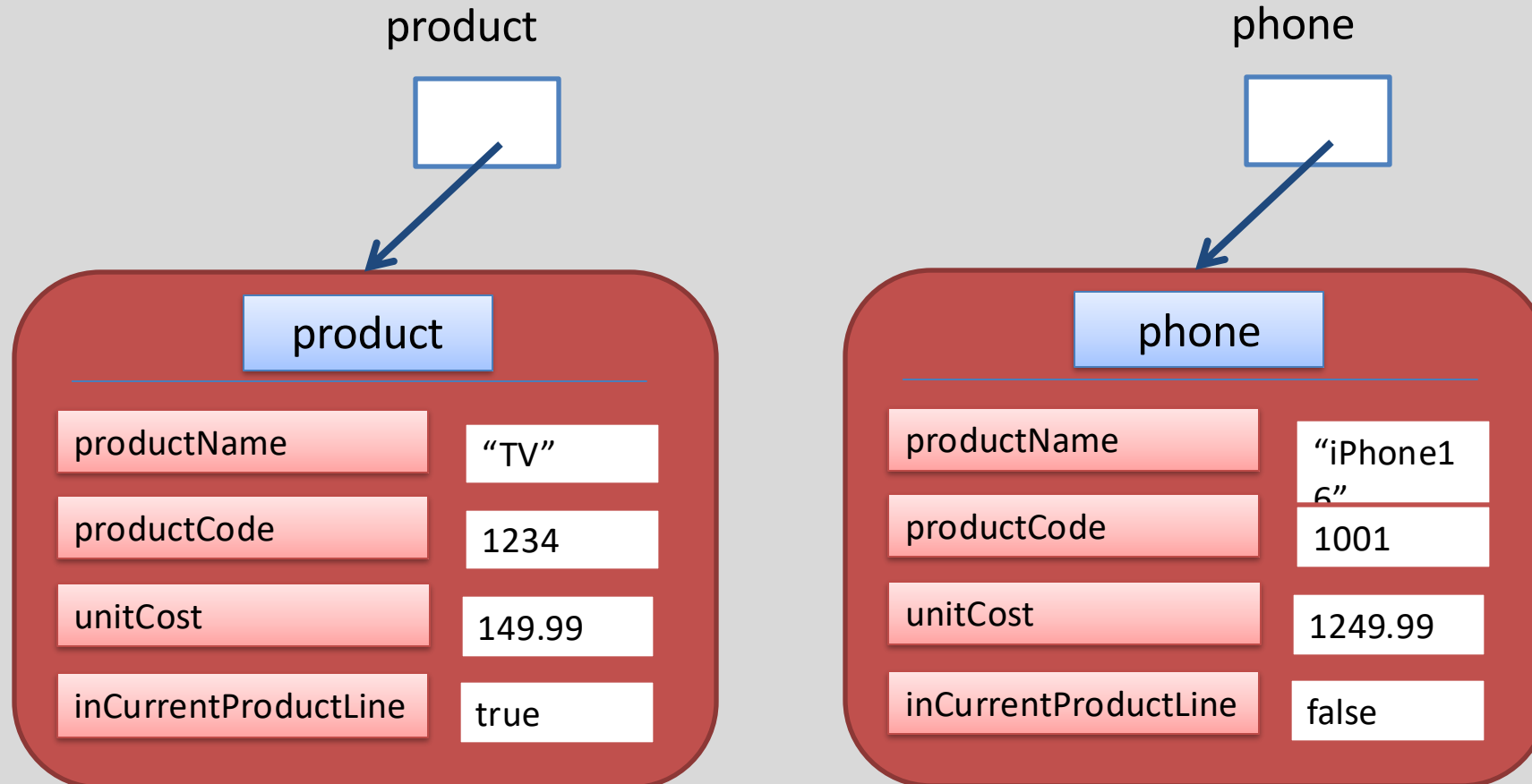
```
private Product product = new Product("TV", 1234, 1149.99, true);
```



Multiple Product Objects

```
private Product product = new Product("TV", 1234, 149.99, true);
```

```
private Product phone = new Product("iPhone16", 1001, 1249.99, false);
```



Questions?

